Solutions

A homogeneous mixture is often referred to as a solution. The continuous medium is the solvent, and the dispersed phase is the solute. The solute disappears after it dissolves, although the color of the solvent may change.

SOLUTIONS

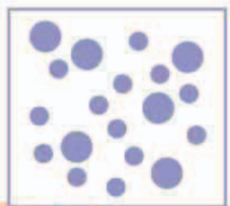
Solvent	Solute	Solution	Description
helium	oxygen	deep-sea breathing gas	helium replaces other gases in the air
air	water	humid air	occurs on warm but damp days
air	smoke	smog	air pollution
water	carbon dioxide	soda water	fizzy water used in sodas
water	ethanoic acid	vinegar	sharp-tasting cooking ingredient
water	salt	seawater	salty water
palladium	hydrogen	palladium hydride	high-tech alloy used in industry
silver	mercury	amalgam	soft alloy used in dental fillings
iron	carbon	steel	high-strength alloy used in construction

Key

Solid Liquid Gas

Suspensions

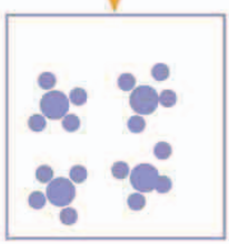
A common type of heterogeneous mixture is a suspension. In contrast to solutions, where the solute breaks up into tiny particles, the particles of the dispersed phase are considerably larger than those of the continuous medium—at least one micrometer across. Everyday examples include the dust carried in wind, tiny droplets in the gas of an aerosol spray, or silt in river water.



▽ **Hanging around**
The particles of the dispersed phase are suspended—they are too small to sink quickly. There are three ways that the mixture can separate.

△ **Creaming**
If the suspended particles are less dense than the continuous phase, they will float. The particles will sit at the surface like cream floating on top of a cup of coffee.

△ **Sedimentation**
If the particles are denser than the continuous phase, they will sink. The particles will form a sediment, or layer, at the bottom of the mixture.



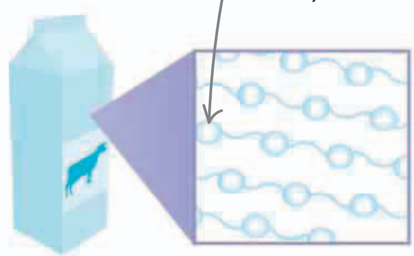
△ **Flocculation**
Sometimes the particles will clump to form larger particles, or floccs. Flocculation happens when the conditions change, or another substance is added to the mixture.

Colloids

A colloid is a mixture that is halfway between a solution and a suspension. The dispersed phase appears to be evenly distributed to the naked eye, but at a microscopic level the two constituents remain heterogeneously mixed. Ice cream, fog, and milk are examples of colloids.

▷ **Cloud**
A cloud is a colloid of liquid water droplets mixed into air. If the droplets grow beyond a certain size, they fall as rain.

fat and water are immiscible (they won't mix), so the fat forms tiny blobs



△ **Milk**
Milk is a colloid of fat in water. Colloids are often white, because the larger size of the dispersed phase causes light to scatter when it passes through the mixture.